Part 0 Introduction

This report aims to propose an adequate linear model to predict the response variable: resale_price of flats in Singapore based on the possible regressors: region, block_group, flat_type, storey, floor_area, flat_model_group, and flat_age (99 - remaining_lease in years).

Singapore housing locations are grouped into regions: Core Central Region (CCR), Rest of Central Region (RCR), and Outside Central Region (OCR), which I used to replace *town* and *street_name* to capture the nature of the value of housing in different regions. Furthermore, since there are 1187 unique blocks in 2456 observations which is too unique, I grouped *blocks* into block_groups based on the hundreds place to better model the area and era they were built. I also grouped storey_range into storey (low, mid, high) and flat_model into flat_model_group to reduce the number of categories. Additionally, I used flat_age to replace *lease_commence_date* and *remaining_lease* to better model the flat age. The variable *month* is fixed in July 2020 which doesn't allow modelling resale price in relation to month and will not be included in the model.

resale_price, floor_area, flat_age are continuous quantitative variables. Region, flat_type, storey, and flat_model_group are ordinal categorical variables. Block is a nominal categorical variable.

I begin with exploratory data analysis to summarise the data and explore the relationships between the variables using summary statistics and visualisations. I will check if the model assumptions are violated through residual analysis. I will transform the response variable and/or regressors if linearity, normality, or homoscedasticity is violated and use WLS if homoscedasticity is violated or GLS if errors are correlated. I will check for multicollinearity by calculating VIF and deal with it using ridge regression or variable elimination. I will use backward elimination for variable selection using the AIC criterion. Finally, I will check that the final model doesn't violate model assumptions.

Part 1 EDA: Exploring the variables and association

Fig 1.1 Histogram of resale_price

The min, lower quartile, upper quartile, and max of resale_price are 160000, 325750, 518000, 1085000 respectively. The mean (436487) is higher than the median (413000), suggesting that the distribution of resale_price is right-skewed and not normal. The standard deviation of resale_price is 152382.4.

From Fig 1.1, the data is mostly clustered around 200000 to 600000. The distribution of resale_price is right_skewed.

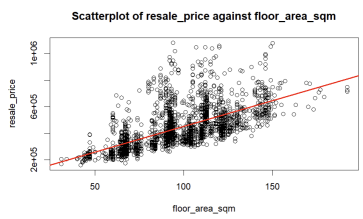

Fig 1.2 Scatterplot of resale_price against floor_area_sqm

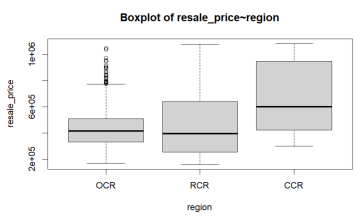

Fig 1.4 Boxplot of resale_price~region

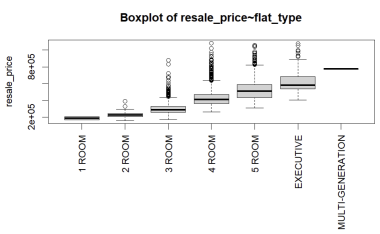

Fig 1.6 Boxplot of resale_price~flat-type

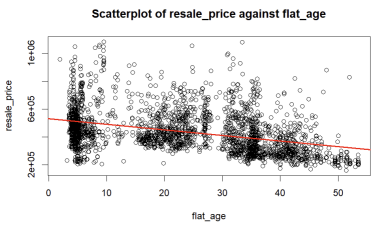

Fig 1.3 Scatterplot of resale_price against flat_age

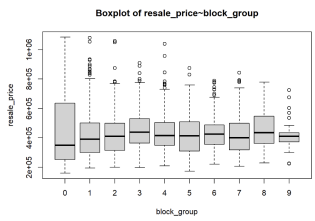

Fig 1.5 Box plot of resale_price~block_group

Fig 1.7 Boxplot of resale_price~storey

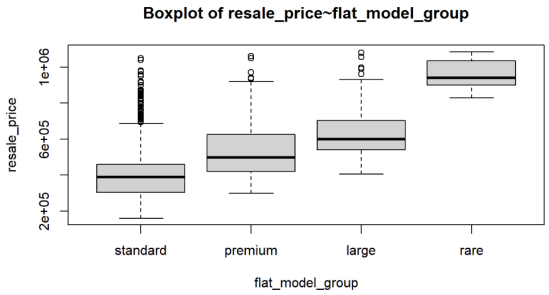

Fig 1.8 Boxplot of resale_price~flat_model_group

It is unsuitable to fit a linear regression model for resale_price without transforming it. From fig 1.1, the distribution of resale_price is asymmetrical and violates the normality assumption.

There is a moderate positive linear relationship between resale_price and floor_area_sqm (cor = 0.638) and a weak negative linear relationship between resale_price and flat_age (cor = -0.379). Fig 1.2 and 1.3 also show signs of non-linearity and significant heteroscedasticity where price variance increases with floor area.

Fig 1.4-1.8 suggest that the distribution of resale_price is heavily right-skewed with numerous upper-end outliers and the constant variance assumption is violated as seen from the non-constant variance across different categories in the boxplots.

Boxcox suggests $\lambda = -0.505 \approx 0$. Thus, I transformed resale_price to log(resale_price) for a more symmetric distribution of the response and to better satisfy the model assumptions.

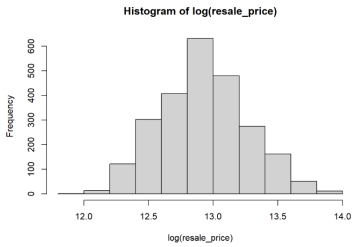

Fig 1.9 Histogram of log(resale_price)

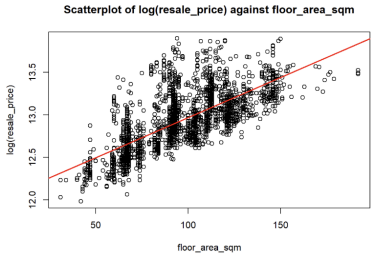

Fig 1.10 Scatterplot of log(resale_price) against floor_area_sqm

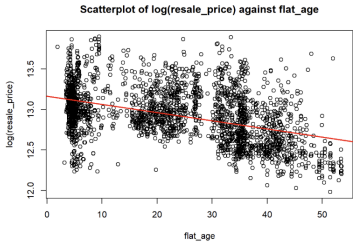

Fig 1.11 Scatterplot of log(resale_price) against flat_age

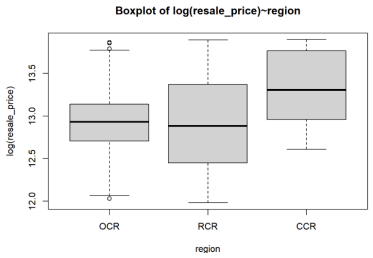

Fig 1.12 Boxplot of log(resale_price)~region

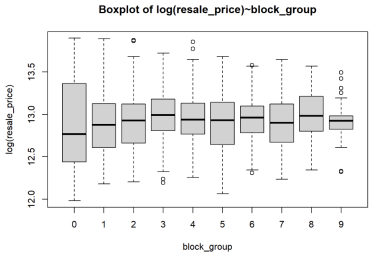

Fig 1.13 Box plot of log(resale_price)~block_group

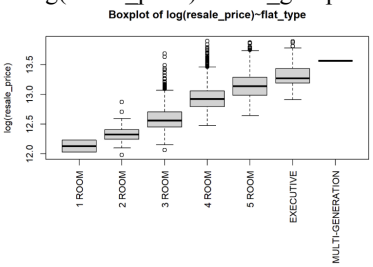

Fig 1.14 Boxplot of log(resale_price)~flat-type

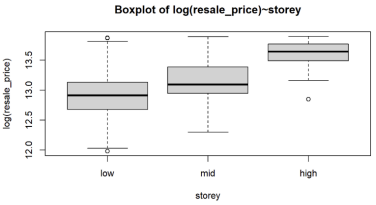

Fig 1.15 Boxplot of log(resale_price)~storey

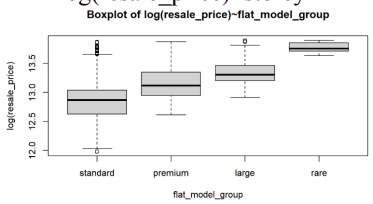

Fig 1.16 Boxplot of log(resale_price)~flat_model_group

From fig 1.9, the distribution of log(resale_price) is more symmetrical and normal. The correlation coefficient between log(resale_price) and floor_area_sqm increased from 0.638 to 0.702 and that between log(resale_price) and flat_age improved from -0.379 to -0.436. Fig 1.10 and 1.11 show a stronger linear relationship between log(resale_price) and floor_area_sqm and flat_age respectively and reduced heteroscedasticity though the variance of log(resale_price) still seems to increase with floor_area_sqm.

Fig 1.12 shows that flats in CCR have higher log(resale_price) than flats in RCR and OCR. Fig 1.13 shows no clear trend between block_group and log(resale_price). Fig 1.14 shows a strong positive relationship between the ordered flat_type and log(resale_price). Fig 1.15 shows a positive relationship between storey and log(resale_price). Fig 1.16 shows a positive relationship between the ordered tiers for flat_model_group and log(resale_price). Fig 1.12 to 1.16 show that constant variance has improved as log(resale_prices) is less unstable across the different categories. WLS or GLS can be used to further stabilise the variance.

Part 2 Building Models (Initial model, M_0)

```
## Coefficients:
##                            Estimate Std. Error t value Pr(>|t|)
## (Intercept)               12.2857108  0.1183593 103.800  < 2e-16 ***
## regionRCR                  0.1712488  0.0133865  12.793  < 2e-16 ***
## regionCCR                  0.3958153  0.0375033  10.554  < 2e-16 ***
## block_group1              -0.0611036  0.0143405  -4.261 2.11e-05 ***
## block_group2              -0.0936745  0.0153751  -6.093 1.29e-09 ***
## block_group3              -0.0801085  0.0161636  -4.956 7.69e-07 ***
## block_group4              -0.1359153  0.0156001  -8.712  < 2e-16 ***
## block_group5              -0.0877165  0.0164464  -5.333 1.05e-07 ***
## block_group6              -0.1493036  0.0160938  -9.277  < 2e-16 ***
## block_group7              -0.0949052  0.0194538  -4.878 1.14e-06 ***
## block_group8              -0.0965353  0.0216205  -4.465 8.37e-06 ***
## block_group9              -0.1430610  0.0278601  -5.135 3.05e-07 ***
## flat_type2 ROOM           -0.0867729  0.1197817  -0.724    0.469
## flat_type3 ROOM            0.0878709  0.1191341   0.738    0.461
## flat_type4 ROOM            0.1519718  0.1223374   1.242    0.214
## flat_type5 ROOM            0.1817066  0.1260890   1.441    0.150
## flat_typeEXECUTIVE         0.0387386  0.1324790   0.292    0.770
## flat_typeMULTI-GENERATION  0.1941734  0.2156913   0.900    0.368
## storeymid                  0.1196198  0.0126838   9.431  < 2e-16 ***
## storeyhigh                 0.4087293  0.0442645   9.234  < 2e-16 ***
## floor_area_sqm             0.0077217  0.0005073  15.221  < 2e-16 ***
## flat_model_grouppremium    0.1084410  0.0110136   9.846  < 2e-16 ***
## flat_model_grouplarge      0.1929422  0.0338769   5.695 1.38e-08 ***
## flat_model_grouprare       0.2655076  0.0662467   4.008 6.31e-05 ***
## flat_age                  -0.0086960  0.0003042 -28.588  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1653 on 2431 degrees of freedom
## Multiple R-squared:  0.7599, Adjusted R-squared:  0.7575
## F-statistic: 320.6 on 24 and 2431 DF,  p-value: < 2.2e-16
```

Fig 2.1 Summary of model1 (initial model)

I fitted log(resale_price) ~ region + block_group + flat_type + storey + floor_area_sqm + flat_model_group +flat_age as shown in fig 2.1.

The fitted model (to 3sf) is $\widehat{log(resale_price)}$ = 12.3 + 0.171*I(region = RCR) + 0.396*I(region = CCR) - 0.0611*I(block group = 1) - 0.0937*I(block group = 2) - 0.0801*I(block group = 3) - 0.136*I(block group = 4) - 0.0877*I(block group = 5) - 0.149*I(block group = 6) - 0.0949*I(block group = 7) - 0.0965*I(block group = 8) - 0.143*I(block group = 9) - 0.0868*I(flat type = 2 room) + 0.0879*I(flat type = 3 room) + 0.152*I(flat type = 4 room) + 0.182*I(flat type = 5 room) + 0.0387*I(flat type = executive) + 0.194*I(flat type = multi-generation) + 0.120*I(storey = mid) + 0.409*I(storey = high) + 0.00772*(floor_area_sqm) + 0.108*I(model = premium) + 0.193*I(model = large) + 0.266*I(model = rare) - 0.00870*(flat_age).

From fig 2.1, at α = 0.05, region, block_group, storey, floor_area_sqm, flat_model_group, flat_age are significant. However, flat_type is not significant, with large p-values (> 0.1). The adjusted R^2 = 0.758, suggesting a good fit. The F-test p-value is < $2.2e^{-16}$ so this model is statistically significant.

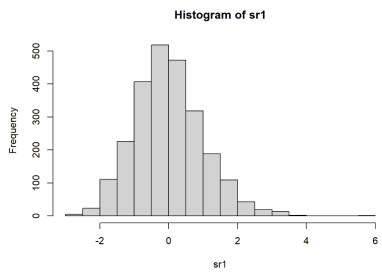

Fig 2.2 Histogram of sr1

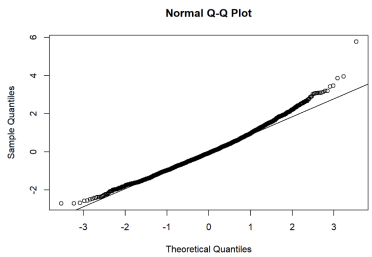

Fig 2.3 QQ plot of sr1

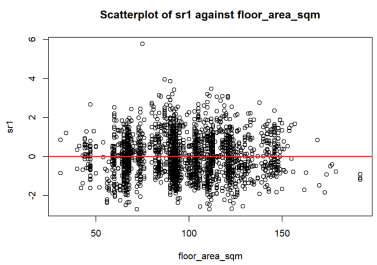

Fig 2.4 Scatterplot of sr1~floor_area_sqm

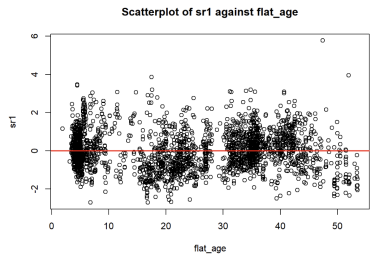

Fig 2.5 Scatterplot of sr1~flat_age

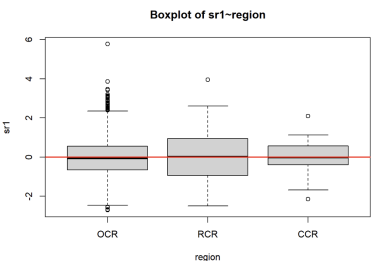

Fig 2.6 Boxplot of sr1~region

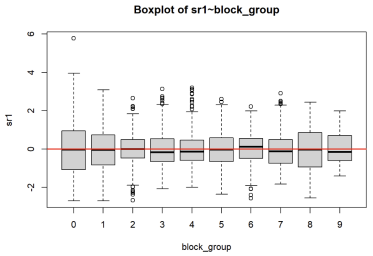

Fig 2.7 Boxplot of sr1~block_group

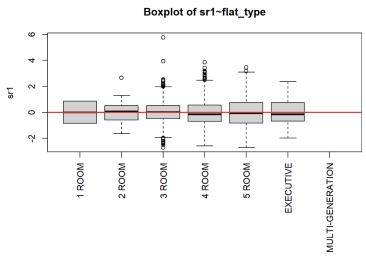

Fig 2.8 Boxplot of sr1~flat_type

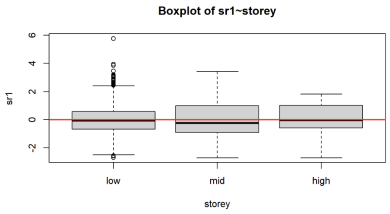

Fig 2.9 Boxplot of sr1~storey

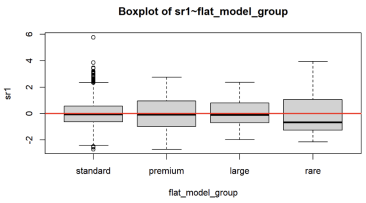

Fig 2.10 Boxplot of sr1~flat_model_group

Fig 2.2 and 2.3 show that the standard residuals are right-skewed and the normality assumption is violated and there might be high-priced outliers. This is confirmed with 16 outliers having |sr1| > 3 corresponding to the extreme points observed in the right tail of the QQ plot. However, since there are no influential points with Cook's distance > 1, we do not need to remove these points. While the heavy right tails suggest a violation of the normality assumption, the large sample size allows the model to be relatively robust due to the Central Limit Theorem.

From fig 2.4 and 2.5, the points are generally randomly scattered around zero without any pattern, suggesting that linearity between log(resale_price) and floor_area_sqm and flat_age is not violated and higher order terms are not required. Homoscedasticity also doesn't seem to be violated. From fig 2.6-2.10, the median line for each category is approximately on the 0 red horizontal line, suggesting that the model is generally unbiased across different categories, supporting the zero-mean error assumption. From fig 2.4 to 2.10, the spread of the residuals is generally consistent across categories, suggesting that constant variance is largely not violated though there's a slight violation of homoscedasticity observed in specific categories. For instance the spread of residuals for categories in block_group, flat_type, and flat_model_group varies. WLS can be used to stabilise the variance.

<u>Part 2 Building Models (Final model, M_N)</u>

I checked for multicollinearity using VIF. GVIF for flat_type is 76.1 which is very large (though its GVIF^(1/(2*Df)) = 1.43 which is small) and GVIF for floor_area_sqm is 14.45 > 10, indicating multicollinearity between flat_type and floor_area_sqm. I decided to drop flat_type from the model to prioritise floor_area_aqm which is a continuous variable that better captures the size of the flat rather than architectural differences in flat_type. This decision is also supported by the fact that flat_type was insignificant in model1. While the adjusted R^2 decreased slightly from 0.758 (initial model, model1) to 0.751 (model2), the resulting model is more parsimonious and now all the GVIFs for the quantitative variables and the GVIF^(1/(2*Df)) for categorical variables < 2, addressing multicollinearity. This also conserved 6 degrees of freedom, further improving the model's efficiency without a substantial loss in explanatory power.

```
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             12.1839859  0.0204386 596.125  < 2e-16 ***
## regionRCR                0.1526815  0.0145625  10.485  < 2e-16 ***
## regionCCR                0.4401132  0.0109502  40.192  < 2e-16 ***
## block_group1            -0.0580794  0.0168701  -3.443 0.000586 ***
## block_group2            -0.0645194  0.0162892  -3.961 7.68e-05 ***
## block_group3            -0.0643299  0.0178644  -3.601 0.000323 ***
## block_group4            -0.1118729  0.0170218  -6.572 6.04e-11 ***
## block_group5            -0.0731091  0.0175931  -4.156 3.36e-05 ***
## block_group6            -0.1265205  0.0166038  -7.620 3.61e-14 ***
## block_group7            -0.1079559  0.0216563  -4.985 6.63e-07 ***
## block_group8            -0.0737502  0.0255955  -2.881 0.003994 **
## block_group9            -0.1374979  0.0253701  -5.420 6.56e-08 ***
## storeymid                0.1090288  0.0135181   8.065 1.13e-15 ***
## storeyhigh               0.4923583  0.0559011   8.808  < 2e-16 ***
## floor_area_sqm           0.0098416  0.0001445  68.131  < 2e-16 ***
## flat_model_grouppremium  0.1200937  0.0127533   9.417  < 2e-16 ***
## flat_model_grouplarge   -0.0210255  0.0153954  -1.366 0.172158
## flat_model_grouprare     0.1701324  0.0955131   1.781 0.074997 .
## flat_age                -0.0074934  0.0002368 -31.649  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.275 on 2437 degrees of freedom
## Multiple R-squared:  0.7845, Adjusted R-squared:  0.7829
## F-statistic: 492.9 on 18 and 2437 DF,  p-value: < 2.2e-16
```

Fig 2.11 Summary of model 3 (WLS)

```
## Analysis of Variance Table
##
## Response: log(resale_price)
##                    Df  Sum Sq Mean Sq  F value    Pr(>F)
## region              2   858.3   429.1  264.075 < 2.2e-16 ***
## block_group         9   293.7    32.6   20.083 < 2.2e-16 ***
## storey              2   666.3   333.2  205.019 < 2.2e-16 ***
## floor_area_sqm      1 10493.5 10493.5 6457.279 < 2.2e-16 ***
## flat_model_group    3   479.6   159.9   98.382 < 2.2e-16 ***
## flat_age            1  1627.8  1627.8 1001.653 < 2.2e-16 ***
## Residuals        2437  3960.3     1.6
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Fig 2.12 Anova table for model 3 (WLS)

Next, I performed Weighted Least Squares (WLS) regression to address heteroscedasticity. From fig 2.11 and 2.12, at $\alpha = 0.05$, all the regressors are still significant in the model. The adjusted R^2 increased from 0.751 (model2) to 0.783 (model3), suggesting a better fit. The F-test p-value is $< 2.2e^{-16}$ so this model is statistically significant.

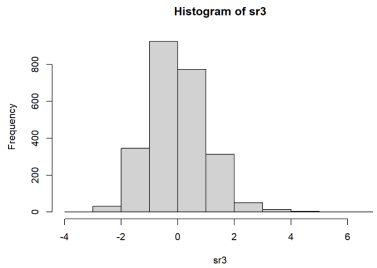

Fig 2.13 Histogram of sr3

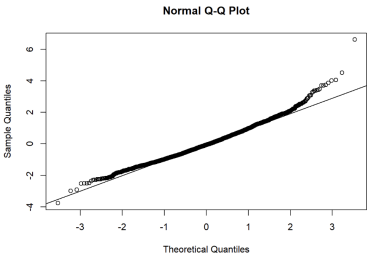

Fig 2.14 QQ plot of sr3

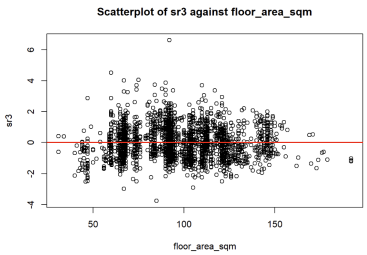

Fig 2.15 Scatterplot of sr3~floor_area_sqm

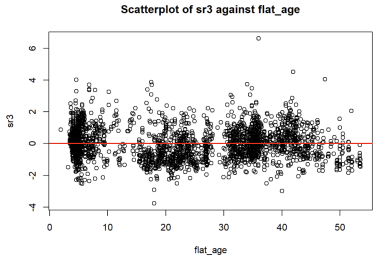

Fig 2.16 Scatterplot of sr3~flat_age

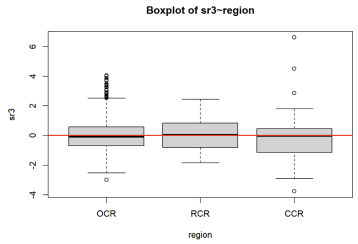

Fig 2.17 Boxplot of sr3~region

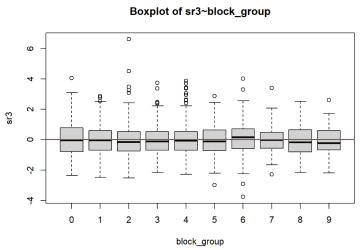

Fig 2.18 Boxplot of sr3~block_group

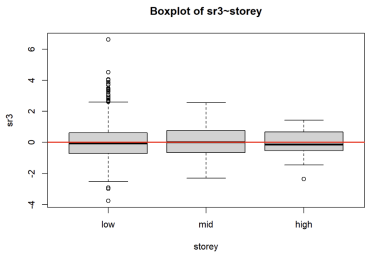

Fig 2.19 Boxplot of sr3~storey

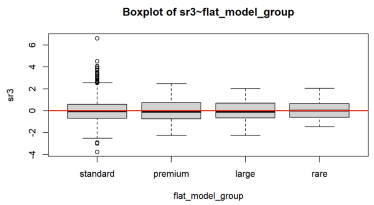

Fig 2.20 Boxplot of sr3~flat_model_group

Fig 2.13 and 2.14 show that the distribution of sr3 is now more normal. The plots for residual analysis from fig 2.14 to 2.20 also suggest improved heteroscedasticity where the spread of the residuals across categories is now more stable and homoscedasticity is not violated.

```
## Start:  AIC=1211.43
## log(resale_price) ~ region + block_group + storey + floor_area_sqm +
##     flat_model_group + flat_age
##
##                     Df Sum of Sq     RSS    AIC
## <none>                           3960.3 1211.4
## - block_group        9     170.3  4130.6 1296.9
## - flat_model_group   3     156.6  4116.8 1300.7
## - storey             2     222.9  4183.2 1341.9
## - flat_age           1    1627.8  5588.0 2055.1
## - region             2    2690.0  6650.2 2480.5
## - floor_area_sqm     1    7543.4 11503.7 3828.4
```

```
## Stepwise Model Path
## Analysis of Deviance Table
##
## Initial Model:
## log(resale_price) ~ region + block_group + storey + floor_area_sqm +
##     flat_model_group + flat_age
##
## Final Model:
## log(resale_price) ~ region + block_group + storey + floor_area_sqm +
##     flat_model_group + flat_age
##
##
##   Step Df Deviance Resid. Df Resid. Dev      AIC
## 1                      2437   3960.288 1211.435
```

Fig 2.21 Results from backward elimination

Finally, I used backward elimination for variable selection based on AIC to reach the most parsimonious model. The final model is the same as model3.

Model3 does not violate normality, linearity, homoscedasticity assumptions and the residuals follow a normal distribution. The observations in the dataset are assumed to be independent, and the model is statistically significant from the F-test. The adjusted R^2 is 0.783 which suggests a good fit. Hence, model3 is a well-fitted and adequate final model to predict resale_price of resale flats in Singapore.

The final fitted model (to 3sf) is $\widehat{log(resale_price)}$ = 12.2 + 0.153*I(region = RCR) + 0.440*I(region = CCR) - 0.0581*I(block group = 1) - 0.0645*I(block group = 2) - 0.0643*I(block group = 3) - 0.112*I(block group = 4) - 0.0731*I(block group = 5) - 0.127*I(block group = 6) - 0.108*I(block group = 7) - 0.0738*I(block group = 8) - 0.137*I(block group = 9) + 0.109*I(storey = mid) + 0.492*I(storey = high) + 0.00984*(floor_area_sqm) + 0.120*I(model = premium) - 0.0210*I(model = large) + 0.170*I(model = rare) - 0.00749*(flat_age).

Holding other regressors constant, an additional square meter leads to an average increase of 0.00984 in log(resale_price) and each year of age reduces log(resale_price) by an average of 0.00749.

Regarding location, with other regressors held constant, flats in from the RCR have an average log(resale_price) that is 0.153 higher than those from the OCR, while flats in the CCR have an average log(resale_price) that is 0.440 higher than the OCR. This suggests a significant price premium for central locations.

With other regressors held constant, block_groups from 1 to 9 have lower average log(resale_price) values compared to block group 0. Block_group 1, 2, 3, 5, and 8 flats show the smallest deviations from block_group 0, with average decreases of log(resale_price) by 0.0581, 0.0645, 0.0643, 0.0731, and 0.0738 respectively. Block_group 4, 6, 7, and 9 have larger average decreases of log(resale_price) by 0.112, 0.127, 0.108 and 0.137 respectively compared to block_group 0.

In terms of floor level, flats on mid storeys have an average log(resale_price) that is 0.109 higher than low storeys, and high storeys have a higher log(resale_price) of 0.492 than low storeys, confirming that higher floors command a price premium, with other regressors held constant.

Finally, relative to the standard flat model group, with other regressors held constant, the premium, and rare categories show higher average log(resale_price) values by 0.120, 0.170 respectively, while large flat model group show lower average log(resale_price) values by 0.0210 (which might be because the effect of size on resale_price is already captured by floor_area_sqm).